NUMERICAL ANALYSIS - LAB WORK 1

Question 5: Performance Comparison & Discussion

To evaluate and compare the performance of Newton's Method, the Secant Method, and Ostrowski's Method, we analyze three primary metrics: the theoretical order of convergence (p), the number of iteration (n), and the overall Efficiency Index (E = p^(1/m)) the function used is $f(x) = x^2 - 2$.

Method	Order of Convergence (p)	Iteration	Efficiency Index (E = p^(1/m))
Secant Method	1.618	7 iterations	$1.618^{1/1}$ ~ **1.618**
Newton's Method	2.0	3 iterations	$2.0^{1/2}$ ~ **1.414**
Ostrowski's Method	4.0	2 iterations	$4.0^{1/3}$ ~ **1.587**

Theoretical Discussion and Trade-offs

The relationship between the computational cost per iteration and the convergence speed determines the overall practical performance of root-finding algorithms:

- **Iterations vs. Evaluation Cost:** Ostrowski's Method achieves fourth-order convergence, requiring the fewest total iterations to reach a given tolerance e. However, it demands 3 function evaluations per step. Newton's Method requires 2 function evaluations per step with quadratic convergence, while the Secant Method requires only 1 new function evaluation per step.
- **Efficiency Index Comparison:** When evaluated purely by the Efficiency Index (E = p^(1/m)), the Secant Method (E ~ 1.618) is theoretically the most efficient per function evaluation, closely followed by Ostrowski's Method (E ~ 1.587), while Newton's Method (E ~ 1.414) is the least efficient.

Practical Advantages and Limitations

- **Secant Method:** *Advantage:* Highly practical when the derivative f' is unknown or computationally expensive to compute. *Limitation:* Lower rate of convergence (p ~1.618) and requires two initial guesses.
- **Newton's Method:** *Advantage:* Simple, standard algorithm with predictable at least quadratic convergence. *Limitation:* Requires evaluation of the derivative f' and fails if f'(x_n) = 0.
- **Ostrowski's Method:** *Advantage:* Rapid convergence to high precision without requiring second derivatives f''. *Limitation:* Higher per-step evaluation cost and risk of numerical instability if f(x_n) - 2f(y_n) ~ 0.